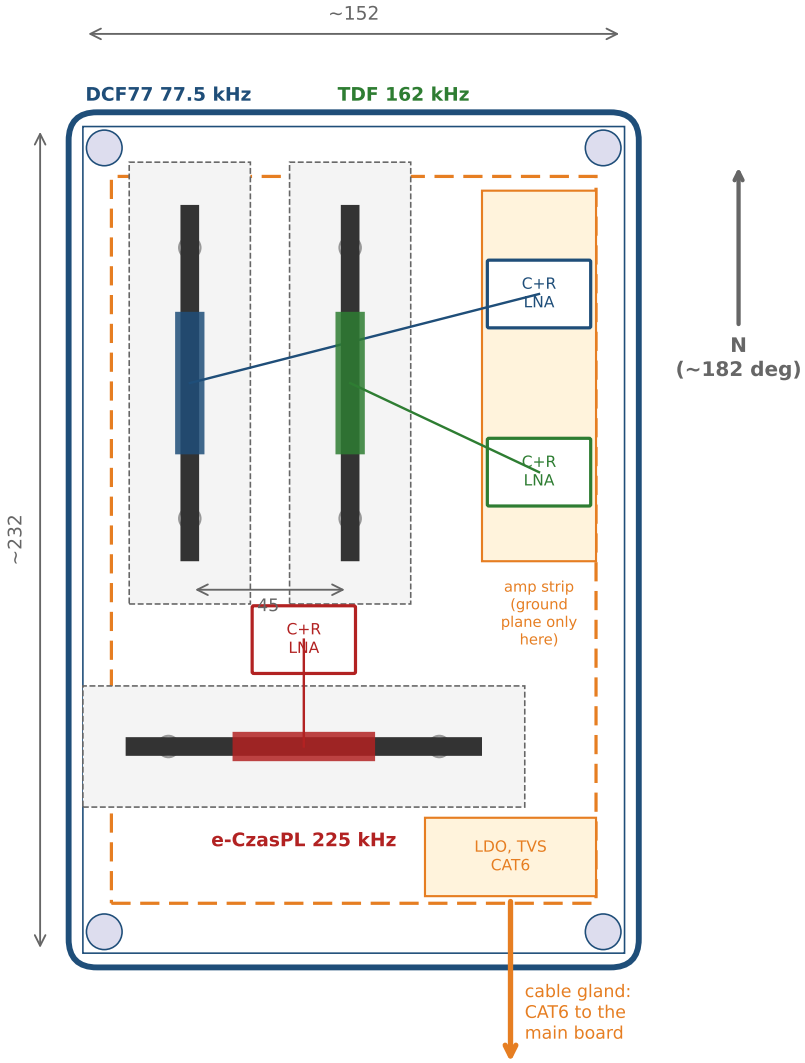
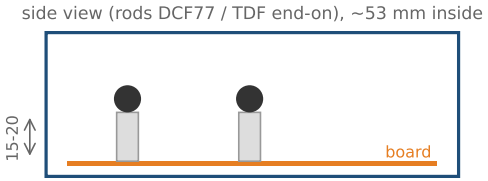
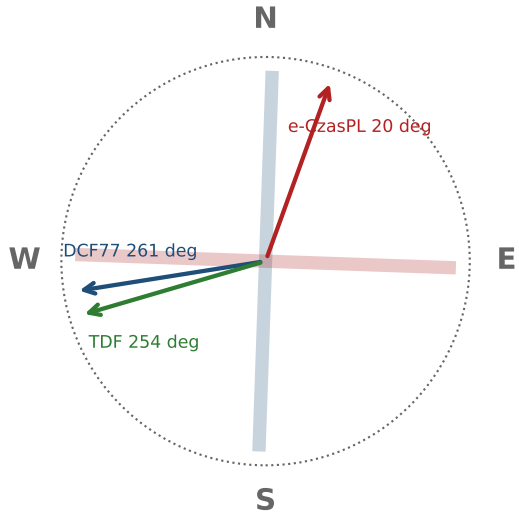
Antenna head - top view to scale (mm)

Hammond 1554VAL2GY (polycarbonate IP68, 240 x 160 x 60 mm outside, ~232 x 152 x 53 inside); rods 10 x 100 mm,
L1 500 uH + 7-turn L2 (rod-antenna-spec.md); head board under the rods; GPS: separate antenna.





station bearings (arrows) and rod axes (bars)



Rods at 90 deg: DCF77 + TDF parallel, e-CzasPL crosswise.

Box long side (DCF77 / TDF axis) to ~182 deg, i.e. ~north-south;
each rod is then off its ideal (broadside) direction by
DCF77 11, TDF 18, e-CzasPL 18 deg: at most -0.45 dB.

Board under the rods, rods on plastic standoffs 15-20 mm above it;
no copper under the rods except narrow traces, ground plane only
in the amp strip and the supply corner, no closed copper loops.

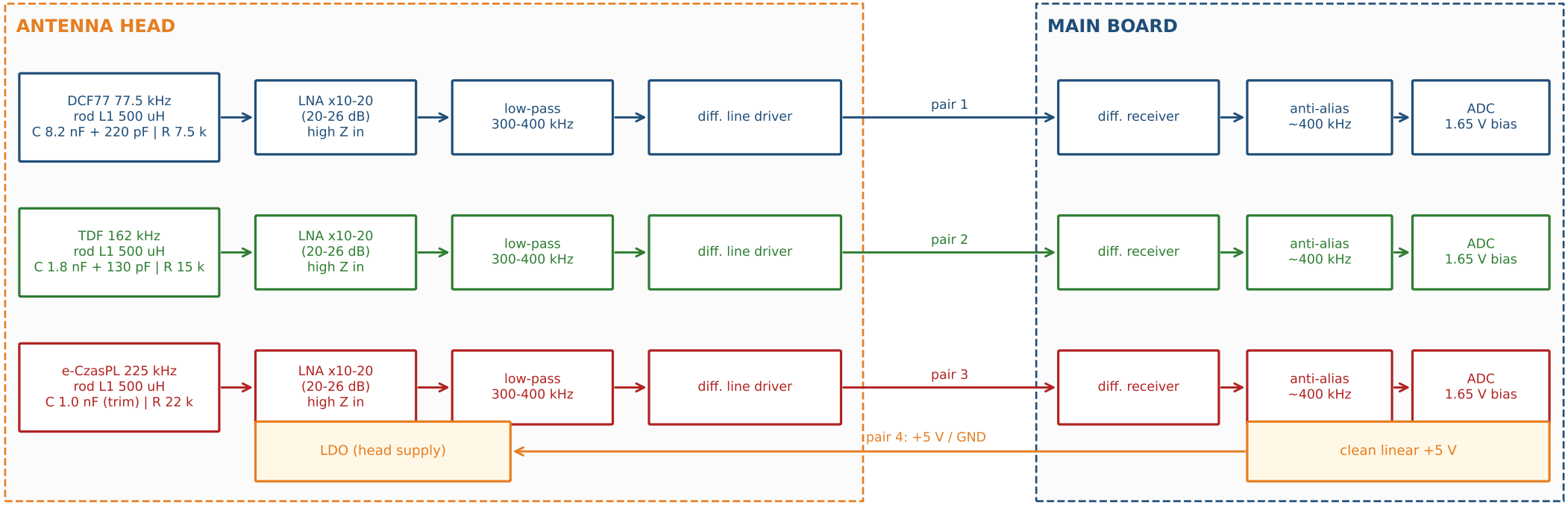
Tank (L1 + C + R, Q ~ 30) at each rod; tune in place with all
three rods mounted and the lid on (SPECTRUM hump).

Rod spacing 45 mm (rods on different frequencies couple weakly):
verify with the prototype (resonance with / without neighbours).

GPS: a separate active antenna (sky view) on its own coax.

150 mm rods: the same layout in a 1554X2GY (300 x 200 x 90).

Antenna head - signal chain per station, cable and main board



CAT5e / CAT6, differential twisted pairs

pair	variant A (three lines)	variant B (one summed line)
1	DCF77	three stations summed
2	TDF	spare
3	e-CzasPL	spare
4	+5 V / GND	+5 V / GND

Values: C0G capacitors at the rod (C = 1/((2 pi f)^2 L), recompute from each rod's measured L), R across L1 for Q ~ 30.

Why the amplifier: the rod's antenna noise must stay well above the ADC noise floor (measured ~15 dB margin at the tank; the 7-turn link alone would cost ~18 dB).

LNA: <= 10 nV/sqrt(Hz), GBW >= 50 MHz for x20 at 225 kHz (or 2 stages).

Variant B: one ADC input for all three stations (STM32H730), separated by the firmware's digital mixers.

ESD/TVS at both cable ends; common-mode rejection by the pairs.

GPS: separate active antenna, own coax to the main board.